Mini Lesson Sample Script

<table>
<tr><td colspan="2">1. Plot data from a table on a line graph to represent change over time.</td></tr>
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<td>The table shows how much Cami weighed at each 4-month period. Plot each point on the coordinate plane, then connect the points with line segments to make a line graph.

<table>
<tr><th>Age (months)</th><th>Weight (pounds)</th></tr>
<tr><td>0</td><td>5</td></tr>
<tr><td>4</td><td>20</td></tr>
<tr><td>8</td><td>30</td></tr>
<tr><td>12</td><td>40</td></tr>
<tr><td>16</td><td>45</td></tr>
<tr><td>20</td><td>50</td></tr>
<tr><td>24</td><td>50</td></tr>
<tr><td>28</td><td>50</td></tr>
</table>
</td>
<td>T: Take a look at this graph. What does the x-axis show?

S: Cami's age in months. It goes up by 4 months each label.

T: And the y-axis?

S: Her weight in pounds. Every line is 5 pounds.

T: Let's plot the first point together. At 0 months, Cami weighed 5 pounds. So I find 0 on the x-axis and go up to 5 on the y-axis. I'll put a point there. Now try the next one. At 4 months, she weighed 20 pounds.

S: I go over to 4 months, then up to 20 pounds. (Plots point.)

T: Plot the rest of the points from the table.

S: (Plots remaining points.)

T: Now connect each point to the next one with a line segment.</td>
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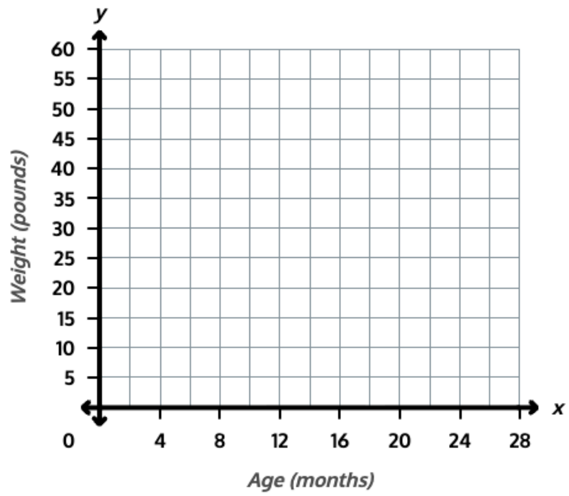
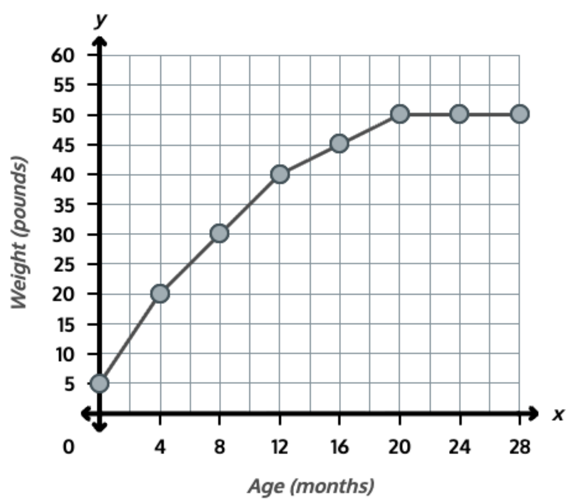


Answer:



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<td>
S: (Draws segments.)

T: What does the line tell you about Cami's weight?

S: She gained weight for a while, and then her weight stopped changing.
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<td colspan="2">2. Analyze the line graph to find differences between data points and compare rates of change.</td>
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Use Cami's weight graph from Problem 1 to answer each question.

a. How much weight did Cami gain in the first 4 months of her life?
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T: How much did Cami weigh at 0 months and at 4 months?

S: 5 pounds, then 20 pounds.
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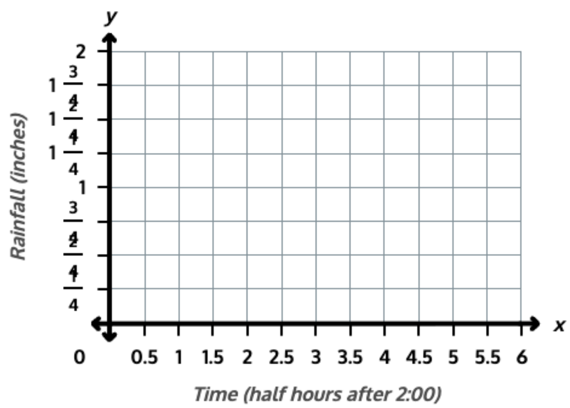
b. How much weight did Cami gain between 4 months and 8 months? c. Did Cami gain more weight in the first 4 months or the second 4 months? How does the line graph show this? *Answer: a) 15 pounds; b) 10 pounds; c) The first 4 months; the line segment from 0 to 4 months is steeper than the segment from 4 to 8 months.*	T: How can you find how much weight she gained? S: I subtract. 20 minus 5 is 15. She gained 15 pounds. T: Now find the gain between 4 and 8 months. S: 30 minus 20 is 10. She gained 10 pounds. T: Did she gain more in the first 4 months or the second 4 months? S: The first 4 months, because 15 is more than 10. T: Look at the two segments on the graph. What do you notice? S: The first one is steeper. It goes up more. T: Right. The steeper segment shows she gained more weight in the same amount of time.
3. **Plot a new data set on a line graph and interpret trends, including intervals of no change.**	
The table shows the total rainfall measured each half hour during a storm. Plot each point on the coordinate plane and connect them with line segments. Then answer the questions.	T: Plot each point from the table, then connect them. S: (Plots points and draws segments.)

Time	Total Rainfall (inches)
2:00	0
2:30	1/4
3:00	3/4
3:30	1
4:00	1
4:30	1
5:00	1 3/4



a. During which half hour did the most rain fall? How can you tell from the graph?

b. What happened to the total rainfall between 3:30 and 4:30? How does the graph show this?

Answer: a) Between 4:30 and 5:00; this segment is the steepest, and the rainfall increased by 3/4 inch (1 3/4 − 1 = 3/4); b) The total rainfall stayed at

T: Look at all the segments. Which one is steepest?

S: The last one, between 4:30 and 5:00.

T: How much rain fell during that half hour?

S: It went from 1 inch to 1 and 3/4 inches. 1 3/4 minus 1 is 3/4 inch.

T: So when a segment is steeper, what does it tell you?

S: More rain fell during that time.

T: Now look at the line between 3:30 and 4:30. What do you see?

S: The line is flat. It stays at 1 inch.

T: What does that tell you about the rain?

S: No rain fell during that hour. The total stayed the same.

1 inch, so no rain fell; the line is horizontal between those times.	